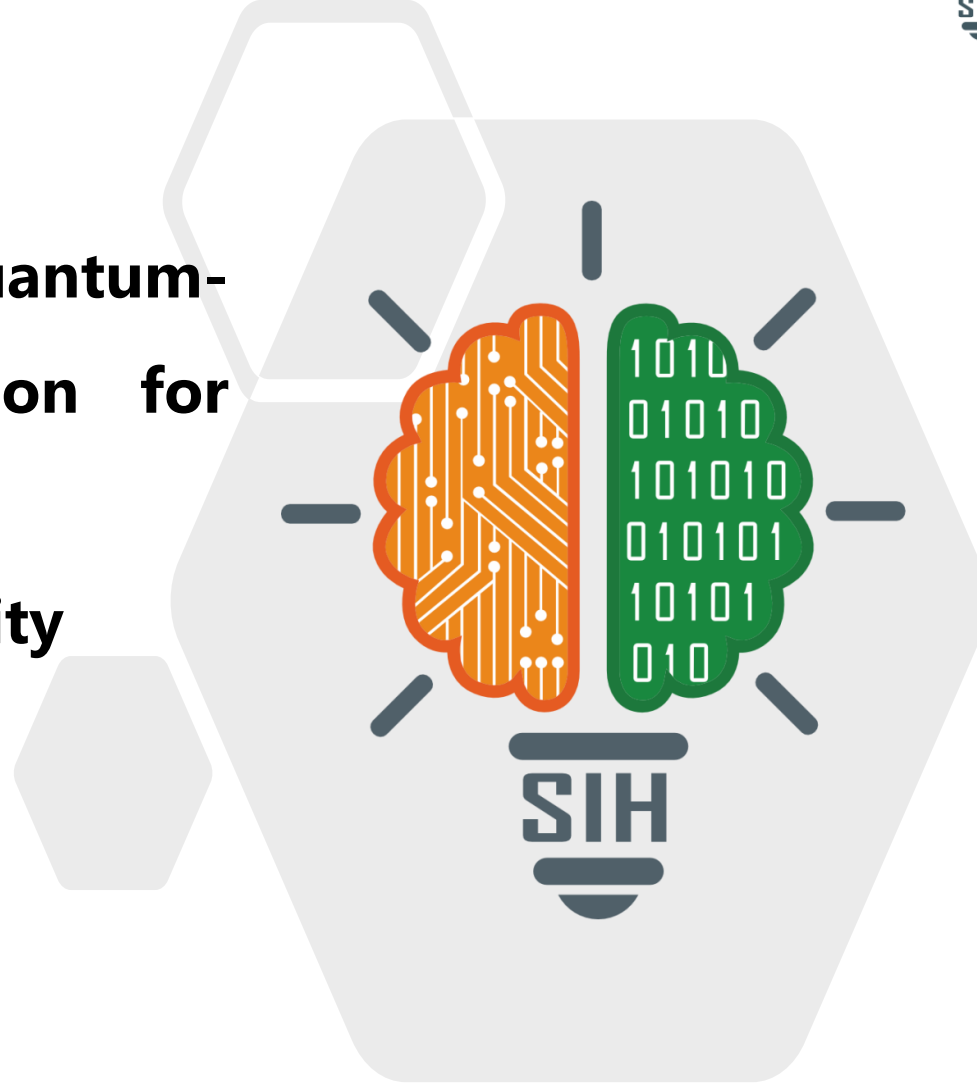


SMART INDIA HACKATHON 2026



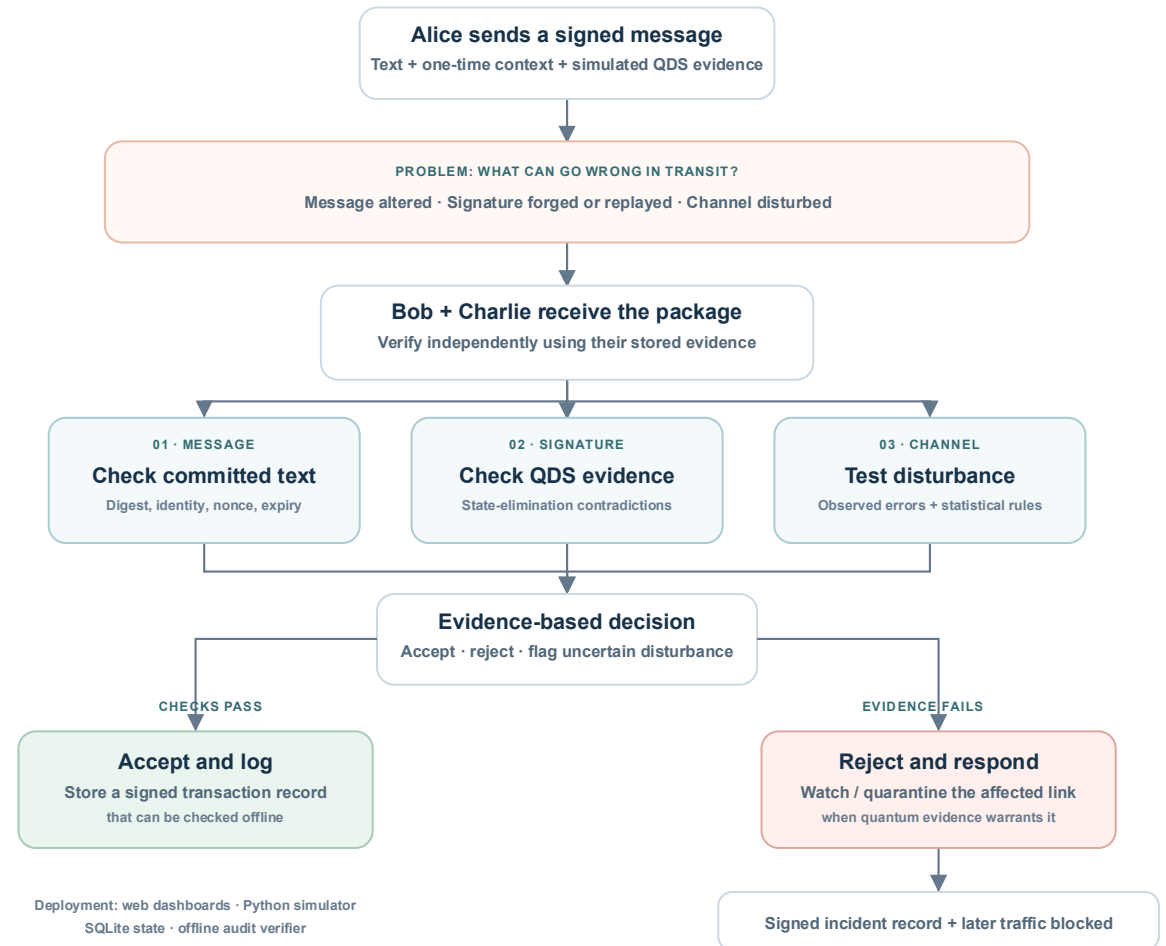
- **Problem Statement ID – 26141**
- **Problem Statement Title- Quantum-Inspired Cyber Threat Detection for Digital Signature Security**
- **Theme- Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID- 159801**
- **Team Name- TEAM ATHENA**





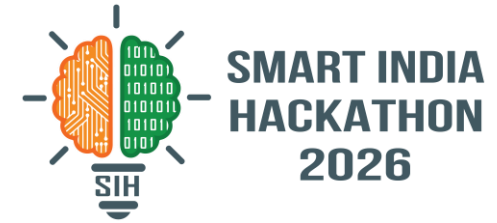
PROPOSED SOLUTION

- Teleportation-Based QDS: Bell-state teleportation + Pauli corrections for secure signature transmission.
- Quantum Threat Detection: Pauli measurements, state elimination, QBER and statistical thresholds detect attacks.
- Message & Session Binding: Hash commitments protect against tampering, replay and impersonation.
- Two-Tier Detection: Distinguishes channel attacks, signer manipulation and uncertain disturbances.
- Quantum Circuit Breaker: WATCH, QUARANTINE, early abort and adaptive testing respond to threats.
- Verifiable Audit Trail: Hash-chained records with optional Ed25519 + ML-DSA-65 hybrid signing.
- Early Attack Detection: Sequential testing can stop transmission before the full signature stream is sent.
- Quantum Circuit Breaker: Suspicious links move through OPEN, WATCH, QUARANTINED and PROBATION states.
- Adaptive Security: WATCH and probation states allocate more test tokens for stronger checking.
- Persistent Audit Trail: Transactions, incidents and resets are stored and survive restarts.



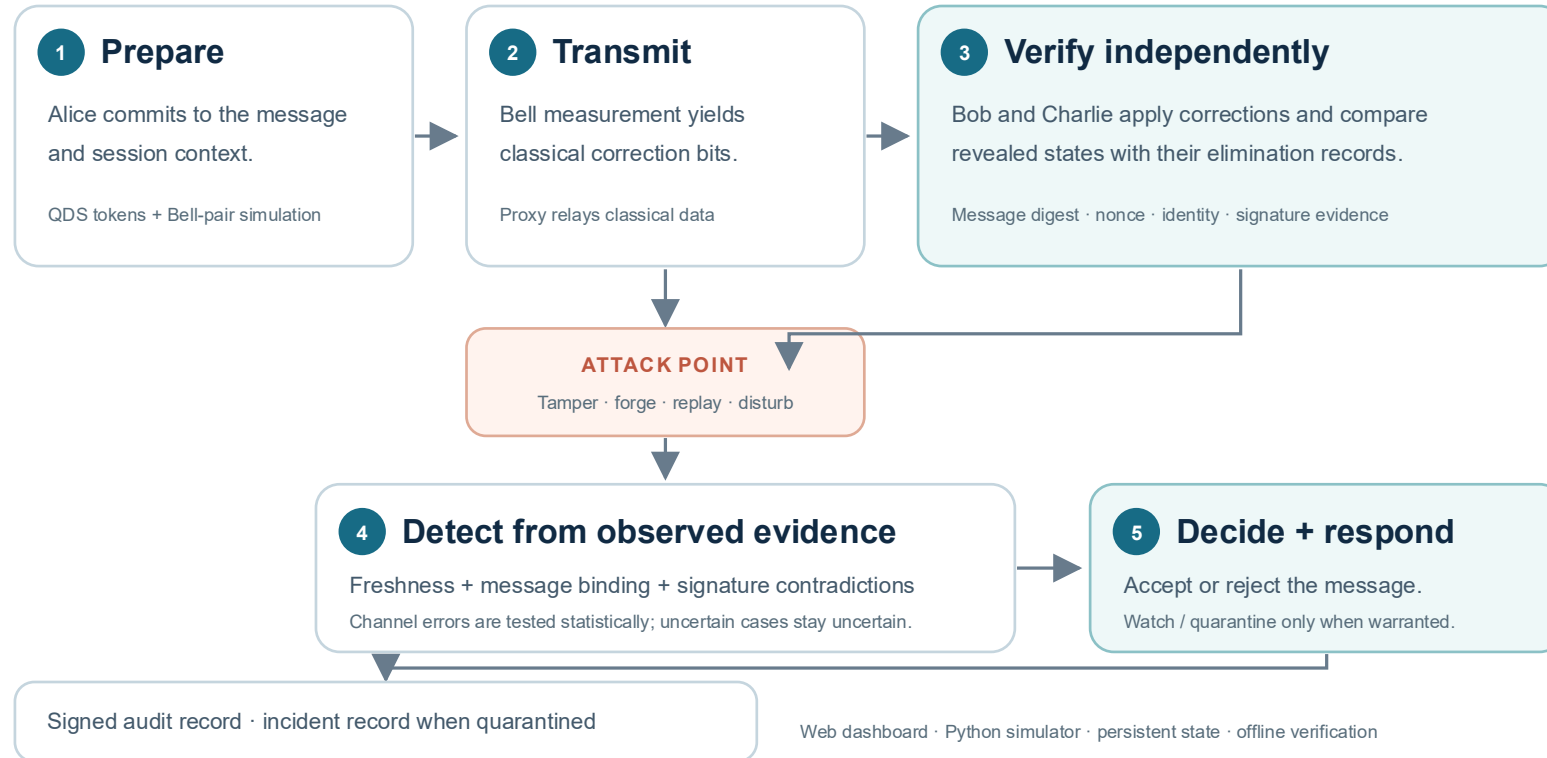


TECHNICAL APPROACH



How the system handles a signed message

Three-party QDS evidence, classical message binding, and an evidence-driven response.



Quantum & Math



Python
3.10+



Qiskit
2.x circuits



Qiskit Aer
AerSimulator



NumPy
QBER metrics



SciPy
Chernoff bounds



Cascade
reconciliation

Network, UI & Testing



asyncio
TCP sockets



JSON
daemon payloads



5 daemons
ports 8000–8004



React.js
live dashboard



HTML5 / CSS
dark theme



Pytest
33 unit tests



FEASIBILITY AND VIABILITY



FEASIBILITY

Hardware Independent Deployment: Capable of executing completely within conventional computer systems through the Qiskit Aer simulator, not requiring any superconducting hardware.

Extremely Efficient Computational Performance: Instant validation possible with $O(1)$ table lookups and deterministic rules.

RUNS ON STANDARD HARDWARE



React.js dashboard live attack telemetry



Deterministic threat engine $O(1)$ table lookups, rule-based



Qiskit Aer simulator quantum circuits and noise models



Conventional computer no superconducting hardware needed

LIKELY CHALLENGES

Decoherence on Channels: Noise from the environment (bit/phase flips) can simulate the effects of an eavesdropper.

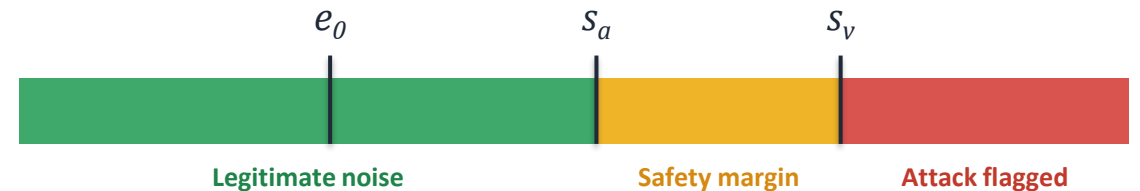
Memory Lifetime: Long-term quantum memory storage is unfeasible on classical technology and machinery.

PROBABLE SOLUTIONS

Threshold Bound Values: Mathematical bounds that define tolerances for legitimate noise and attacks ($e_0 < s_a < s_v$)

Immediate Projective Measurement: Tokens are measured as soon as they are received, hence no quantum memory is required.

NOISE vs. ATTACK THRESHOLDS



mismatch rate (QBER) increases $\rightarrow$



IMPACT AND BENEFITS



IMPACTS

Defense & Strategic Sectors: Neutralizes "Harvest Now, Decrypt Later" threats against classified military communications.

FinTech & Banking: Secures high-value transactions with non-repudiable, quantum-proof validation.

Government Infrastructure: Provides tamper-proof authentication for national identity and critical document verification.



Defense

Blocks "harvest now, decrypt later"



FinTech & Banking

Non-repudiable, quantum-proof transactions



Government

Tamper-proof identity and documents

BENEFITS

Information-Theoretic Security: Protection backed by physical laws (No-Cloning Theorem), not computational complexity.

Explainable & Auditable: Eliminates black-box AI bias and hallucinations, meeting rigorous regulatory compliance standards.

Zero GPU Overhead: Minimal energy and compute footprint compared to large-scale machine learning threat models.

	❌ Conventional	✅ Our approach
Security basis	Math hardness that Shor's algorithm can break	Physical laws (No-Cloning Theorem)
Decisions	Black-box ML, false positives	Deterministic and auditable
Compute cost	GPU-heavy threat models	Zero GPU overhead



RESEARCH AND REFERENCES



FOUNDATIONAL RESEARCH

Wallden et al. (2015): [Quantum digital signatures with quantum key distribution components](#), Phys. Rev. A / arXiv:1403.5551.

Gottesman & Chuang (2001): [Quantum Digital Signatures](#) (arXiv:quant-ph/0105032).

Wang et al. (2015): [Quantum Digital Signature Scheme based on Multiparty Controlled EPR Channels](#) (arXiv:1507.03581).

Amiri et al. (2015): [Practical Quantum Digital Signature without authenticated quantum channels](#) (arXiv:1507.03333)

CORE PRINCIPLES WE BUILD ON



No-Cloning Theorem
unknown states cannot be copied



Uncertainty Principle
eavesdropping leaves a trace



Quantum Teleportation
Bell pairs + Pauli feed-forward



Chernoff-Hoeffding
finite-sample error bounds

RESEARCH LINEAGE

